

Lesson 10: Partial Order Relations

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Partial Orderings

Definition: A relation R on a set S is called a **partial ordering** or **partial order** if it is reflexive, anti-symmetric, and transitive. A set S together with a partial ordering R is called a **partially ordered set**, or **poset**, and is denoted by (S, R) . Members of S are called elements of the poset.

Example -1: Show that the “greater than or equal” relation ($\geq$) is a partial ordering on the set of integers. That is; $(\mathbb{Z}, \geq)$ is a poset.

- Since $a \geq a, \forall a \in \mathbb{Z}$, $\geq$ is reflexive.
- If $a \geq b$ and $b \geq a$, then $a = b$. Hence $\geq$ is anti-symmetric.
- If $a \geq b$ and $b \geq c$, then $a \geq c$. Hence $\geq$ is transitive.

Since $\geq$ is reflexive, anti-symmetric, and transitive, $\geq$ is a partial order relation on the set of integers and $(\mathbb{Z}, \geq)$ is a poset.

More Examples

Example - 2: Show that the divisibility relation $|$ is a partial ordering on the set of positive integers. That is; $(\mathbb{Z}^+, |)$ is a poset.

- Since $a|a, \forall a \in \mathbb{Z}^+, |$ is reflexive.
- If $a|b$ and $b|a$, then $a = b$. Hence $|$ is anti-symmetric.
- If $a|b$ and $b|c$, then $a|c$. Hence $|$ is transitive.

Since $|$ is reflexive, anti-symmetric, and transitive, $|$ is a partial order relation on the set of positive integers and $(\mathbb{Z}^+, |)$ is a poset.

More Examples

Example -3: Show that the inclusion relation $\subseteq$ is a partial ordering on the power set of a set S . That is; $(\wp(S), \subseteq)$ is a poset.

- Since $A \subseteq A, \forall A \in \wp(S), \subseteq$ is reflexive.
- If $A \subseteq B$ and $B \subseteq A$, then $A = B$. Hence $\subseteq$ is anti-symmetric.
- If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. Hence $\subseteq$ is transitive.

Since $\subseteq$ is reflexive, anti-symmetric, and transitive, $\subseteq$ is a partial order relation on $\wp(S)$ and $(\wp(S), \subseteq)$ is a poset.

Exercise

Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings? Justify your answer.

1. $R_1 = \{(0, 0), (1, 1), (2, 2), (3, 3)\}$
2. $R_2 = \{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
3. $R_3 = \{(0, 0), (1, 1), (1, 2), (2, 2), (3, 3)\}$
4. $R_4 = \{(0, 0), (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$
5. $R_5 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$.

Solution:

1. R_1 is a partial ordering since it is reflexive, anti-symmetric, and transitive.
2. R_2 is not a partial ordering since it is not anti-symmetric $((2, 3), (3, 2) \in R_2$ but $2 \neq 3)$.
3. R_3 is a partial ordering since it is reflexive, anti-symmetric, and transitive.
4. R_4 is a partial ordering since it is reflexive, anti-symmetric, and transitive.
5. R_5 is not a partial ordering since it is not transitive $((2, 0), (0, 1) \in R_5$ but $(2, 1) \notin R_5)$.

Example -4

Is (S, R) a poset if S is the set of all people in the world and $(a, b) \in R$, where a and b are people, if

1. a is taller than b ? No.
2. a is not taller than b ? No.
3. $a = b$ or a is an ancestor of b ? Yes.
4. a and b have a common friend? No.

Totally Ordered Set

Definition: The elements a and b of a poset $(S, \preceq)$ are called **comparable** if either $a \preceq b$ or $b \preceq a$. When a and b are elements of S such that neither $a \preceq b$ nor $b \preceq a$, a and b are called **incomparable**.

Example: In the poset $(\mathbb{Z}^+, |)$, are the integers 3 and 9 comparable? Are 5 and 7 comparable?

Definition: If $(S, \preceq)$ is a poset and every two elements of S are comparable, S is called a **totally ordered** or **linearly ordered** set, and $\preceq$ is called a **total order** or a **linear order**. A totally ordered set is also called a **chain**.

Example: The poset $(\mathbb{Z}, \geq)$ is totally ordered, because $a \geq b$ or $b \geq a$ whenever a and b are integers.

Example: The poset $(\mathbb{Z}^+, |)$ is not totally ordered because it contains elements that are incomparable, such as 5 and 7.

Well-ordered Set

Definition: Let $(S, \preceq)$ be a poset and let A be a non-empty subset of S .

An element $l \in A$ is said to be the **least element** of A if and only if $l \preceq x, \forall x \in A$.

An element $g \in A$ is said to be the **greatest element** of A if and only if $x \preceq g, \forall x \in A$.

Definition: $(S, \preceq)$ is a **well-ordered** set if it is a poset such that $\preceq$ is a total ordering and every nonempty subset of S has a least element.

Example: The poset $(\mathbb{Z}^+, \leq)$ is a well-ordered set.

Example: The poset $(\mathbb{Z}, \leq)$ is not a well-ordered set.

The Principle of Well-Ordered Induction

Theorem: Suppose that $(S, \preccurlyeq)$ is a well-ordered set and $P(x)$ is a propositional function over the domain S . If for every $y \in S$, if $P(x)$ is true for all $x \in S$ with $x \preccurlyeq y$, then $P(y)$ is true, then $P(x)$ is true for all $x \in S$.

$$[\forall y \in S (\forall x \in S [(x \preccurlyeq y \wedge P(x) \text{ is true}) \rightarrow P(y) \text{ is true}])] \rightarrow \forall x \in S [P(x) \text{ is true}].$$

Proof: Suppose that $P(x)$ is not true for all $x \in S$. Then the set $A = \{x \in S \mid P(x) \text{ is not true}\} \neq \varnothing$.

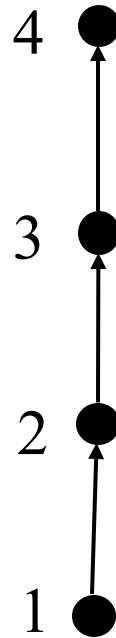
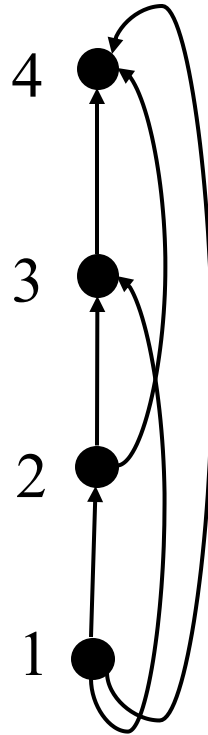
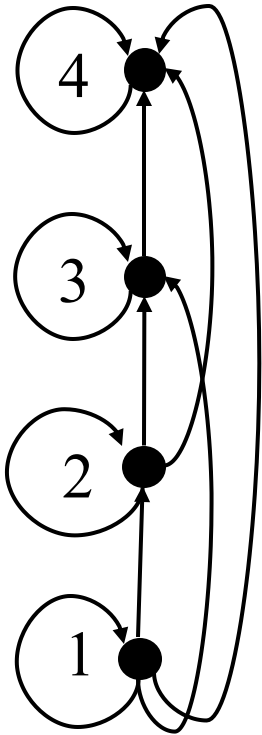
Because S is well-ordered, A has a least element a . By the choice of a as a least element of A , we know that $P(x)$ is true for all $x \in S$ with $x \preccurlyeq a$.

Therefore, by the hypotheses of the theorem, $P(a)$ is true. This is a contradiction.

Hence, $P(x)$ must be true for all $x \in S$.

Hasse Diagrams

Let $S = \{1, 2, 3, 4\}$. Draw the directed graph for the partial ordering $\{(a, b) \mid a \leq b\}$ on S .

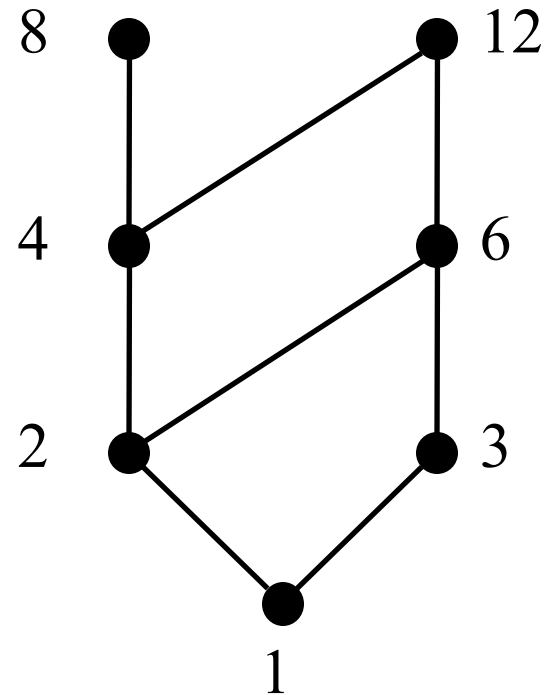


1. Remove all the loops.
2. Remove all edges (x, y) for which there is an element $z \in A$ such that $x < z$ and $z < x$.
3. Remove all the arrows on the directed edges, because all edges point “upward” toward their terminal vertex.

The resulting diagram is called the Hasse diagram of $(S, \leq)$, named after the twentieth-century German mathematician Helmut Hasse.

Hasse Diagrams

Example 5: Draw the Hasse diagram representing the partial ordering $\{(a, b) | a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.



Maximal and Minimal Elements of a Poset

Definition: An element $a \in S$ is **maximal** in the poset $(S, \preceq)$ if there is no $b \in S$ such that $a \preceq b$.

Similarly, an element $a \in S$ is **minimal** if there is no $b \in S$ such that $b \preceq a$.

An element $a \in S$ is the **greatest element** of the poset $(S, \preceq)$ if $b \preceq a$ for all $b \in S$.

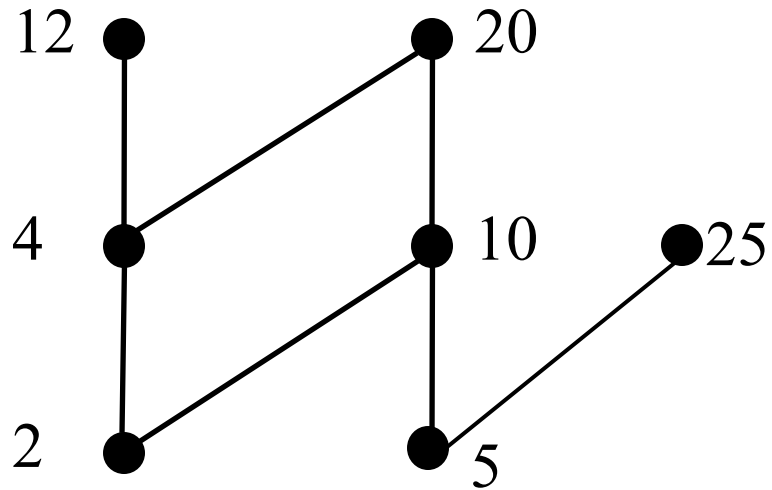
Similarly, an element $a \in S$ the **least element** of the poset $(S, \preceq)$ if $a \preceq b$ for all $b \in S$.

Example: Let $S = \{2, 4, 5, 10, 12, 20, 25\}$. Which elements of the poset $(S, |)$ are maximal, and which are minimal?

Example: Is there a greatest element and a least element in the poset $(\mathbb{Z}^+, |)$?

Example 6

Let $S = \{2, 4, 5, 10, 12, 20, 25\}$. Which elements of the poset $(S, |)$ are maximal, and which are minimal?



The Hasse Diagram of a Poset $(S, |)$

Maximal elements of $S = \{12, 20, 25\}$

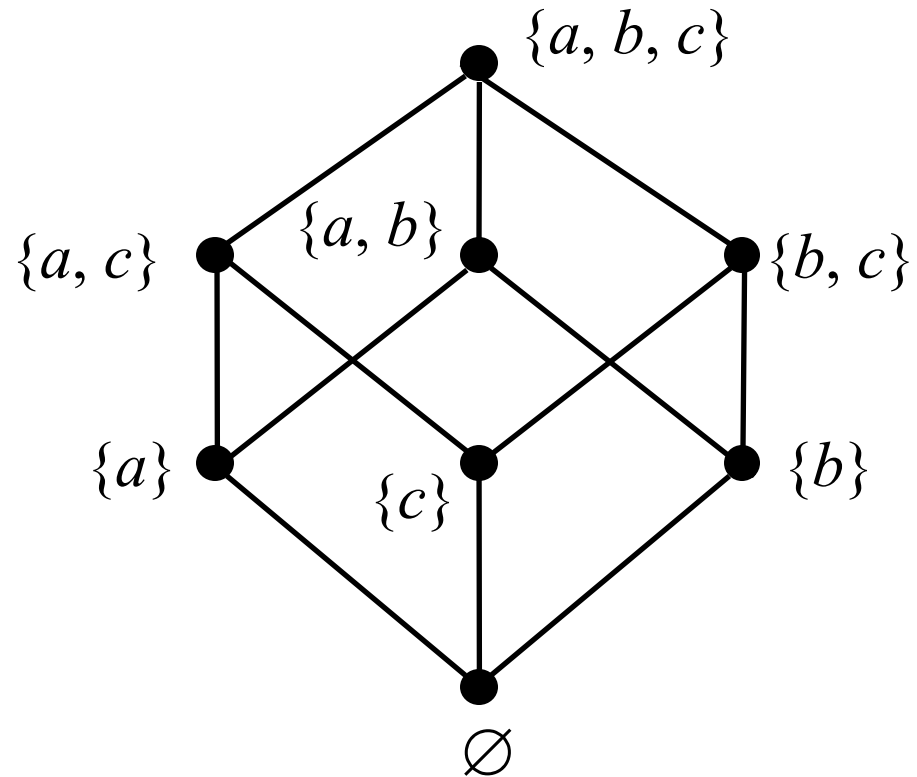
Minimal elements of $S = \{2, 5\}$

There is no least element for S

There is no greatest element for S .

Hasse Diagrams

Let $S = \{a, b, c\}$. Draw the Hasse diagram for the poset $(\wp(S), \subseteq)$.



Least element of $\wp(S)$ is φ

Greatest element of $\wp(S)$ is $S = \{a, b, c\}$

Upper and Lower bounds of a subset of a Poset

Definition: Let $(S, \preceq)$ be a poset and let $A \subseteq S$. If u is an element of S such that $a \preceq u$ for all elements $a \in A$, then u is called an **upper bound** of A .

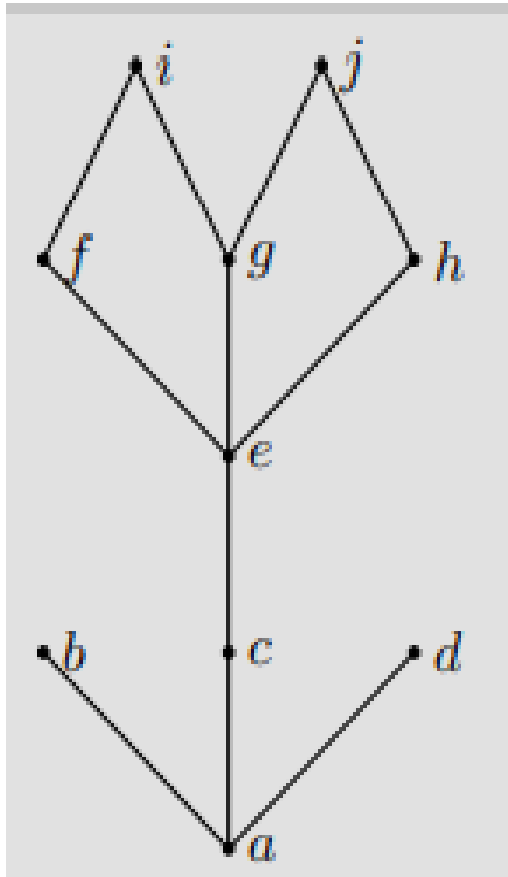
An element $x \in S$ that is an upper bound of A and $x \preceq u$ for all upper bounds u of A is called the **least upper bound** (lub) of A .

Definition: Let $(S, \preceq)$ be a poset and let $A \subseteq S$. If l is an element of S such that $l \preceq a$ for all elements $a \in A$, then l is called a **lower bound** of A .

An element $x \in S$ that is a lower bound of A and $l \preceq x$ for all lower bounds l of A is called the **greatest lower bound** (glb) of A .

Example 7

Consider the Hasse diagram of the poset $(S, \leq)$.



Minimal elements of $S = \{a\}$.

Maximal elements of $S = \{b, d, i, j\}$.

Upper bounds of $\{c, e\} = \{e, f, g, h, i, j\}$.

Least upper bound (lub) of $\{c, e\} = e$.

Lower Bounds of $\{c, e\} = \{a, c\}$.

Greatest lower bound (glb) of $\{c, e\} = c$.

Find the greatest lower bound and the least upper bound of $\{b, i\}$, if they exist.

$glb(\{b, i\}) = a$.

$lub(\{b, i\})$ does not exist.

Example 8

Find the greatest lower bound and the least upper bound of the sets $\{3, 9, 12\}$ and $\{1, 2, 4, 5, 10\}$, if they exist, in the poset $(\mathbb{Z}^+, |)$.

Solution: Lower bounds of $\{3, 9, 12\}$ are 1 and 3.
Since $1 \mid 3$, 3 is the greatest lower bound of $\{3, 9, 12\}$.

The only lower bound of $\{1, 2, 4, 5, 10\}$ is 1. Hence the greatest lower bound of $\{1, 2, 4, 5, 10\}$ is 1.

An integer is an upper bound for $\{3, 9, 12\}$ if and only if it is divisible by 3, 9, and 12.
The integers with this property are those divisible by the least common multiple of 3, 9, and 12, which is 36.
Hence, 36 is the least upper bound of $\{3, 9, 12\}$.

The least common multiple of 1, 2, 4, 5, 10 is 20.
Hence, 20 is the least upper bound of $\{1, 2, 4, 5, 10\}$.